

Posting messages in mid-air

Dying to make known your view on the last film you saw or the last book you read? If you have your mobile at hand, you may soon be able to post



your views right outside your front door. Researchers the world over are working on ways to create mid-air messaging systems that let you post or read comments that people have left tied to a particular location. The 'mid-air' messages will hang in the air until someone walks past carrying a device, a phone, handheld computer or laptop—any device capable of receiving the messages.

The uses are manifold. Sailors could be informed about the latest hazards or other dangers with the use of mid-air messaging systems. Waterfront tours in New York use mid-air messaging to inform the audience of information at every nook and corner. One of the bigger mid-air messaging projects taking place in Bristol has just received a grant from the government to fund its development.

However, a big problem is identifying the right 'locating system' to use. GPS, which uses signals from satellites, only works well outside and its co-ordinate system is only able to resolve areas down to a few meters in diameter. The other problem is working out which messages to serve up to people as they pass. The software loaded on the devices will have to know about its user's likes and dislikes, so it only gives them a notice of the things they are interested in.

Master key with 'oracle attack'

A computer security expert at A&T Labs in New Jersey, US, Matt Blaze, has detailed how to make a master key for an entire office building or apartment block using only one sample key and a simple process of trial and error. The applied technique is known as an 'oracle attack'. This is used to quickly shortlist the solutions to a cryptography problem, when some knowledge of the key has been gained.

The technique works with pin-and-tumbler locks for which a master key can be designed to open many different doors. This is the most commonly used type of lock for large buildings. An attacker needs only one sample key, access to its lock and a bag of blank keys ready for cutting. Pin-and-tumbler locks only open when a key has elevated all the pins inside the mechanism to their correct positions.

A normal lock with five

pins can have around a million different pin configurations. However, locks for which there is a master key have two sets of correct pin positions. The first is unique to each individual lock; the second works with the master key and is the same for every lock. In a document submitted to the journal *IEEE Security and Privacy*, Blaze shows that it is possible to work out the master key posi-

tions in a relatively short amount of time using the method of trial and error.

Locksmiths say that they are well aware of the trick. But Blaze says he is the first to set out a mathematical theory underpinning it in detail.

Even though it is technically simple and could be used by criminals, he argues it is necessary to publicise the loophole. Blaze says many professional security consultants are unaware of the exploit and need to be told.



quoteworthy

"We still strongly believe that digital copyrights should be protected, but mandating a tax is not the right solution"

Chuck Mulloy, spokesman for Intel, on the recent proposal in Germany that PC makers pay copyright holders a 'copyright tax' of 12 Euros for every system sold to compensate for the unauthorised copying of digital movies and other copyrighted material.

"I had one item on my agenda: not to lose. I decided it would be wiser to stop playing"

Garry Kasparov, international chess grandmaster, after he accepted a draw in a six-game chess contest against Deep Junior, the supercomputer that won last year's world computer chess championship. Deep Junior can process 3 million moves in a second and chooses strategy over just capturing pieces.

tomorrow's technology

Mind the chip

In what may be the most advanced project of its kind, researchers at Infineon Technologies (www.infineon.com) and the Max Planck Institute (<http://www.mpi.nl/world/index.html>) have succeeded in getting actual communication between live brain cells and a tiny solid state semiconductor.

Here is where the neurological monitoring of tomorrow comes in: a tiny 1 square

millimeter chip containing some 16,364 sensors that detect the faint pulses given out by neurons.

The weak electrical impulses emitted by neurons are amplified and transferred over to a computer for processing. With highly sensitive sensors in place, neurobiologists can analyse how associated groups of neurons or individual cells respond to electrical stimulation or spe-

cific substances over a defined period of time. The tissue and the chip are dipped into an electrolytic nutrient fluid that keeps the cells alive.

The neuro-chip lets scientists understand the neurobiological and neurochemical functioning of nerve tissue over a sustained period of time. It is mainly used in the research of cures for diseases of the brain, such as Alzheimer's disease.